Machine Learning for Professional Chess Match Outcome Prediction

Conor Jack Vincent

University of Lincoln, School of Computer Science

26071875@students.lincoln.ac.uk

Abstract. This work presents a literature review on sport prediction using machine learning techniques, exploring the key themes, challenges, and research gaps. Subsequently, a project progress update is provided, illustrating both completed and ongoing objectives with reference to an updated Project Plan and Gant chart.

Keywords: Machine Learning, Chess Prediction, Predictive Modeling, Data Collection, Data Preprocessing

1 Literature Review

Recently, the application of machine learning techniques in predicting sport match outcomes has gained significant attention. Both team sports and individual sports have been of interest, and research has explored the large magnitude of features that could influence results. Bunker and Susnjak (2022) carried out a comprehensive review of 31 papers focused on predicting match results within different team sports, published between 1996 to 2019. This review highlighted common methodologies used across each reviewed paper, such as accuracy being a common metric to determine model performance and Artificial Neural Networks (ANNs) being the commonly used modelling technique. In conjunction to this, the authors note that the lack of standardized benchmark datasets and differences in sport characteristics makes comparisons between different studies difficult. Additionally, the paper illustrated the importance of feature selection and engineering and noted that greater focus should be placed on this phase of the process to achieve superior outcomes.

While Bunker and Susnjak's review primarily focuses on various team sports, the fundamental principles relate to the challenges encountered when predicting outcomes of professional chess matches. Echoing these principles, a review by Haghighat et al. (2013) on Data Mining Techniques for Result Prediction in Sports discusses similar concepts. The study explores the application of data mining and modelling in assisting coaches and managers for result prediction in various sports while recognizing the challenges that arise during data collection and feature selection. Again, the authors of this review highlight the need for a standardized dataset, acknowledging the current need to collect data from different sources leading to inconsistent datasets.

Within the realm of team sports, Hucaljuk and Rakipovic (2011) tackled the task of classifying football match outcomes as either a win, loss, or draw. The authors touched upon the difficulty in predicting football games due to the multitude of factors requiring consideration, some of which may be challenging to quantify. Like the previous studies, challenges were faced when collecting historical data, leading to the reliance on the current year's league statistics. Various machine learning models, including Naïve Bayes, Bayesian Networks, Logit Boost, Random Forests, and ANNs were used, with ANNs outperforming the rest achieving an accuracy of 68%. This model showed promise however it was noted future improvements within feature selection could lead to increased performance.

Similarly, Huan and Chang (2010) made a study specifically focused on predicting football game outcomes during the 2006 World Cup, adopting a neural network method based on a Multi-Layer Perceptron (MLP) with back propagation. The authors used official statistical data from previous stages of the tournament to predict the winning rates of two teams. The model managed to obtain a prediction accuracy of 76.9%, excluding drawn games. Again, it was noted the difficulty involved in prediction due to the different situations and factors causing unpredictability within football. However, this paper illustrated the significance of determining optimal model parameters.

In relation to outcome prediction for individual sports, Somboonphokkanphan et al. (2009) carried out a study using Time-Series History with Neural modelling to predict winners within Tennis. This study showcases three distinct MLP models, each employing different sets of input parameters which helped highlight the influence of feature selection on outcome predictions. The MLP model, trained with appropriate features which focused on a player's experience over the past year and included environment details such as type of tennis court, demonstrated superior accuracy compared to existing tennis match prediction models.

Transitioning to the domain of chess, Reddy et al. (2023) performed chess match outcome predictions through sequential data analysis coupled with deep learning. This study utilized a small dataset culminating in 324 players with varying ratings collected from 'Chess.com' and 'Lichess.org' and employed Recurrent Neural Networks and Long Short-Term Memory Networks for binary and multiclass classification tasks achieving an accuracy of 99.2%. Additionally, Pulido (2016) utilized both statistical and machine learning techniques to predict chess game outcomes after twenty moves, employing the Gorgo Base dataset. Two models were utilized, Support Vector Machine (SVM) and Random Forest, and both showed reasonable results, though data limitations were acknowledged. Another study carried out by Thabtah et al. (2010) focused on analyzing the impact of player ratings (Elo) on outcome predictions, showing that incorporation of player ratings improves model accuracy. Unlike the first two mentioned chess studies, this project will focus on utilizing pregame statistics including player ratings alongside providing a larger dataset in terms of sample size and number of features.

In conclusion, while predictive modeling in sports outcomes has been explored thoroughly, the application to chess, especially targeted towards titled players, remains unexplored. This proposed project aims to fill this gap, alongside creating a

comprehensive dataset that will be made publicly available to help tackle the difficulties that arise in obtaining standardized datasets for cross study comparisons.

2 Progress Update

2.1 Completed Tasks

In terms of achieving a machine learning model with the ability to predict professional chess match outcomes, progress has been made in several tasks illustrated in the project plan (see Table 1). Task 1.1 has been undertaken requiring a programming language and set of libraries to be chosen for the project. It's important to note that the libraries chosen so far are not exhaustive and will be subject to change throughout the project dependent on arising requirements. The selected programming language is Python, with this decision being influenced by the library options available for data handling, manipulation, and machine learning. Additionally, the Application Programming Interfaces (APIs) selected for data collection are accessible through Python wrapper libraries.

To obtain a feature set the Python libraries 'Pandas', 'Json', 'Urllib', and 'Pickle' where chosen. 'Pandas' was incorporated to enable the creation of datasets from collected data and facilitate any data manipulation or analysis (Pandas, 2024). The 'Json' library was selected as the API requests returned data in the JavaScript Object Notation (JSON) format and needed decoding into Python datatypes such as dictionaries or lists (Python Software Foundation, 2024. a). The 'Urllib' library was utilized as some JSON API responses included Uniform Resource Locators (URLs) that when opened contained further data that could be collected (Python Software Foundation, 2024. b). Lastly in terms of data handling, the 'Pickle' library is used to store collected data in a 'Pandas' dataframe, enabling permanent storage and eliminating the need for reacquiring data (Python Software Foundation, 2024. c). Finally, concerning the machine learning aspect of the project, three libraries have been identified, these are 'scikit-learn', 'TensorFlow' and 'PyTorch'. Each of these libraries can be utilized for model creation and will be selected depending on the machine learning model requirements (Scikit-learn, 2024; TensorFlow, n.d.; PyTorch n.d.).

Objective 2.1 and 2.2 have been completed, which came with the requirement of finding appropriate APIs for collecting relevant chess data (see Table 1). The two APIs that were found and considered are the 'Chess.com' API and 'Lichess.org' API (Bcurtis, 2022; Lichess, n.d.). Both APIs have Python wrapper libraries facilitating requests being made through Python code (Chess.com, n.d.). In this instance, the API being used is currently the 'Chess.com' API with the 'chessdotcom' library. The 'Chess.com' API enables data to be collected for all listed titled players along with player's statistics for each of the 'Chess.com' game types and furthermore access to each of the players game history.

2.2 Tasks in Progress

At this point in the project, both objectives 3.1 and 3.2 are underway, which are to collect data using the chosen API and transform this data into a structured format. Currently, 9 different 'Pandas' dataframes (structured data) have been created and stored using the 'Pickle' library. The first dataframe created contains the player profiles of 10089 players who hold an official title, including information such as usernames, titles, number of followers, country, paying status, streamer status, verification status and current league. From this dataset, a further 8 datasets were created using the usernames from the first, obtaining statistics for each of the 'Chess.com' game types (see fig 1). Originally, the plan was to collect statistical and game data from 100 different players, specifically Grand Masters, for pairwise matchups between these players. However, after carrying out API requests for each of the titled players and obtaining specific statistics, some of the requests failed, meaning the loss of data samples and in conjunction to this, not all players have data for each collected statistic. Therefore, the sample size was widened by introducing any titled player into the feature set to ensure a result with more samples with the 'required' features.

Fig. 1. This figure shows the tables mentioned in the section 'Tasks in Progress' from this report merged which has a current total of 76 columns and a total of 10089 rows. This dataset illustrates the total amount of data collected so far using the 'Chess.com' API.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10089 entries, 0 to 10088
Data columns (total 76 columns):
 #   Column                                    Non-Null Count  Dtype
---  ------                                    --------------  -----
 0   name                                      7948 non-null   object
 1   username                                  10089 non-null  object
 2   title                                     10089 non-null  object
 3   followers                                 10089 non-null  int64
 4   country_code                              10089 non-null  object
 5   country                                   10089 non-null  object
 6   status                                    10089 non-null  object
 7   is_streamer                               10089 non-null  bool
 8   verified                                  10089 non-null  bool
 9   league                                    8342 non-null   object
 10  chess_daily_last_rating                   4152 non-null   float64
 11  chess_daily_last_date                     4152 non-null   float64
 12  chess_daily_last_rd                       4152 non-null   float64
 13  chess_daily_best_rating                   3400 non-null   float64
 14  chess_daily_best_date                     3400 non-null   float64
 15  chess_daily_best_game                     3400 non-null   object
 16  chess_daily_record_win                    4152 non-null   float64
 17  chess_daily_record_loss                   4152 non-null   float64
 18  chess_daily_record_draw                   4152 non-null   float64
 19  chess_daily_record_time_per_move          4152 non-null   float64
 20  chess_daily_record_timeout_percent        4152 non-null   float64
 21  chess_daily_tournament_points             102 non-null    float64
 22  chess_daily_tournament_withdraw           102 non-null    float64
 23  chess_daily_tournament_count              102 non-null    float64
 24  chess_daily_tournament_highest_finish     102 non-null    float64
 25  chess960_daily_last_rating                898 non-null    float64
 26  chess960_daily_last_date                  898 non-null    float64
 27  chess960_daily_last_rd                    898 non-null    float64
 28  chess960_daily_best_rating                749 non-null    float64
 29  chess960_daily_best_date                  749 non-null    float64
 30  chess960_daily_best_game                  749 non-null    object
 31  chess960_daily_record_win                 898 non-null    float64
 32  chess960_daily_record_loss                898 non-null    float64
 33  chess960_daily_record_draw                898 non-null    float64
 34  chess960_daily_record_time_per_move       898 non-null    float64
 35  chess960_daily_record_timeout_percent     898 non-null    float64
 36  chess960_daily_tournament_points          50 non-null     float64
 37  chess960_daily_tournament_withdraw        50 non-null     float64
 38  chess960_daily_tournament_count           50 non-null     float64
 39  chess960_daily_tournament_highest_finish  50 non-null     float64
 40  chess_rapid_last_rating                   7676 non-null   float64
 41  chess_rapid_last_date                     7676 non-null   float64
 42  chess_rapid_last_rd                       7675 non-null   float64
 43  chess_rapid_best_rating                   7359 non-null   float64
 44  chess_rapid_best_date                     7359 non-null   float64
 45  chess_rapid_best_game                     7359 non-null   object
 46  chess_rapid_record_win                    7676 non-null   float64
 47  chess_rapid_record_loss                   7676 non-null   float64
 48  chess_rapid_record_draw                   7676 non-null   float64
 49  chess_bullet_last_rating                  8357 non-null   float64
 50  chess_bullet_last_date                    8357 non-null   float64
 51  chess_bullet_last_rd                      8357 non-null   float64
 52  chess_bullet_best_rating                  8112 non-null   float64
 53  chess_bullet_best_date                    8112 non-null   float64
 54  chess_bullet_best_game                    8112 non-null   object
 55  chess_bullet_record_win                   8357 non-null   float64
 56  chess_bullet_record_loss                  8357 non-null   float64
 57  chess_bullet_record_draw                  8357 non-null   float64
 58  chess_blitz_last_rating                   9743 non-null   float64
 59  chess_blitz_last_date                     9743 non-null   float64
 60  chess_blitz_last_rd                       9743 non-null   float64
 61  chess_blitz_best_rating                   9705 non-null   float64
 62  chess_blitz_best_date                     9705 non-null   float64
 63  chess_blitz_best_game                     9705 non-null   object
 64  chess_blitz_record_win                    9743 non-null   float64
 65  chess_blitz_record_loss                   9743 non-null   float64
 66  chess_blitz_record_draw                   9743 non-null   float64
 67  fide                                      5541 non-null   float64
 68  tactics_highest_rating                    10089 non-null  int64
 69  tactics_highest_date                      10089 non-null  int64
 70  tactics_lowest_rating                     10089 non-null  int64
 71  tactics_lowest_date                       10089 non-null  int64
 72  puzzle_rush_best_total_attempts           8080 non-null   float64
 73  puzzle_rush_best_score                    8080 non-null   float64
 74  puzzle_rush_daily_total_attempts          410 non-null    float64
 75  puzzle_rush_daily_score                   410 non-null    float64
dtypes: bool(2), float64(57), int64(5), object(12)
memory usage: 5.7+ MB
```

2.3 Progress Reflection

The project's current progress is slightly behind schedule, with tasks 3.1 and 3.2 still in progress (see Table 1 and fig 2). The reason for this delay is due to various challenges, including solving 'Chess.com' API errors, handling failed requests, and creating 'Pandas' dataframes from complex nested JSON objects obtained through API requests. Furthermore, the allotted time for these objectives did not sufficiently reflect the actual time needed and ultimately, the complexity of creating a robust labelled dataset using an API was underestimated. This delay should not have a significant impact on the overall project's deadline or outcome, and the Project Plan and Gant chart have been modified to account for these delays.

Table 1. This Project Plan is the newly modified plan which has had new Start Dates and End Dates appended in red, considering the current challenges and delays within the project. The old deadlines have been kept highlighting the changes that have been made.

Objective	Aim	Deliverables	Start Date	End Date	No of Weeks
1.1	Determine programming language that will be used and available libraries.	Decision on programming language and selected libraries that will be used to create both a dataset and a machine learning model.	06/11/2023	12/11/2023	1
2.1	Find Application Programming Interfaces (APIs) for data collection.	Identified different API's that allow for the extraction of data from online chess platforms.	13/11/2023	26/11/2023	2
2.2	Investigate the available API's end points and capabilities to gain an understanding of what data can be collected.	Understanding of what data should and can be collected through the previously found APIs.	27/11/2023	10/12/2023	2
3.1	Begin using the APIs to collect any data that is deemed relevant for the dataset that will be used for training the machine learning model.	Semi-structured data acquired using previously found APIs.	11/12/2023	31/12/2023 19/02/2024	3 10
3.2	Turn semi-structured historical chess data into structured data.	Historical chess data in structured format.	01/01/2024 20/02/2024	14/01/2024 26/02/2024	2 1
4.1	Carry out data cleaning to remove any	Obtained cleaned dataset with no	15/01/2024 27/02/2024	28/01/2024 04/03/2024	2 1

	inaccurate or unwanted data within the dataset and merge collected data.	errors ready for use in the creation of the feature set.			
4.2	Carry out data preprocessing and identify features that can be used or created from the structured historical chess dataset.	Created a feature set ready for training a machine learning classification model with the inclusion of labels.	29/01/2024 05/03/2024	19/02/2024 11/03/2024	3 1
5.1	Determine two different supervised machine learning algorithms that can be used in reference to the feature set and output requirements.	Two Machine learning algorithms selected for use.	19/02/2024 12/03/2024	25/02/2024 18/03/2024	1 1
5.2	Train one of the machine learning algorithms on the feature set.	One machine learning model trained on the created feature set.	26/02/2024 19/03/2024	10/03/2024 25/03/2024	2 1
5.3	Train second machine learning algorithm on the feature set.	Both machine learning models trained on the created feature set.	11/03/2024 26/03/2024	24/03/2024 01/04/2024	2 1
6.1	Evaluate Both of the machine learning models using relevant metrics.	Obtained valuable insights that can be used to evaluate both machine learning models performance.	25/03/2024 02/04/2024	07/04/2024 08/04/2024	2 1
6.2	Use acquired insights about both machine learning models to compare the difference in performance.	Identified which model performed better on the chess feature set.	08/04/2024 09/04/2024	14/04/2024 15/04/2024	1 1
6.3	Compare the results of machine learning models against other relevant machine learning models in the realm of sport and chess match outcome prediction.	Identified how well the created model performs with reference to other models within the same research area.	15/04/2024 16/04/2024	28/04/2024 29/04/2024	2 2

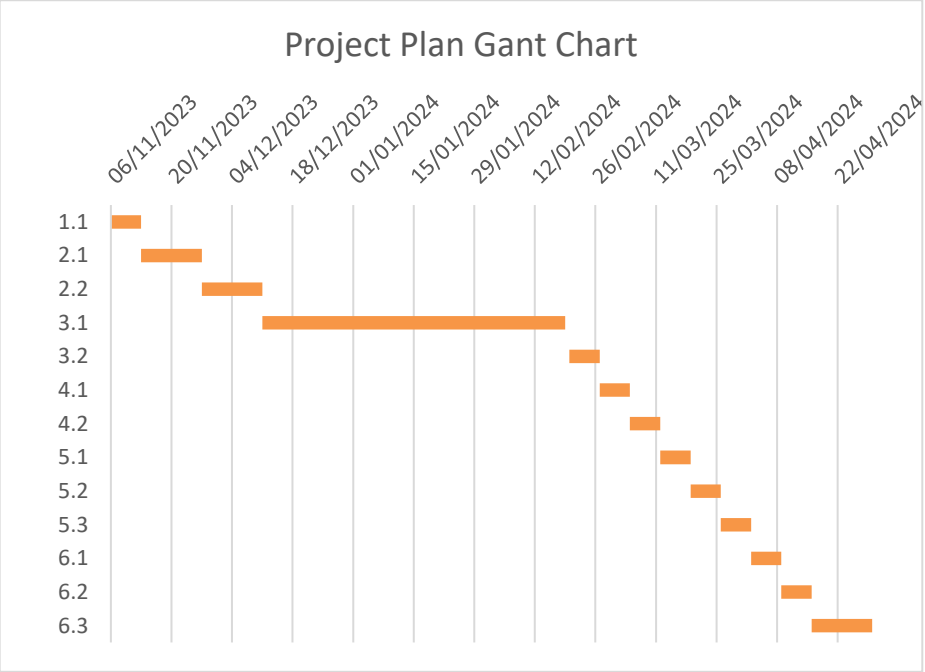

Fig. 2. The above figure is a Gant chart created in excel spreadsheet from the updated project plan table (see **Table 1**). The time scale of each line within the Gant chart is 14 days. For example, objective 1.1 is shown to end in between two date lines, illustrating that the time taken is 1 week or 7 days.

References

Bcurtis (2022) *Published-Data API* [Blog]. 10 May. Available from https://www.chess.com/news/view/published-data-api [accessed 12 January 2024].

Bunker, R. and Susnjak, T. (2022) The Application of Machine Learning Techniques for Predicting Match Results in Team Sport: A Review. *Journal of Artificial Intelligence Research*, 73, 1285-1322. Available from https://www.jair.org/index.php/jair/article/view/13509/26786 [accessed 12 January 2024].

Chess.com (n.d.) *Welcome to Chess.com Wrapper's Documentation!* Chess.com. Available from https://chesscom.readthedocs.io/en/latest/ [accessed 12 January 2024].

Haghighat, M., Rastegari, H. and Nourafza, N. (2013) A Review of Data Mining Techniques for Result Prediction in Sports. *Advances in Computer Science: an International Journal*, 2(5) 7 – 12. Available from https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=37767ab1f559dda7a8d239aec0af1c9485d3d426 [accessed 13 January 2024].

Huang, K. and Chang, W. (2010) A neural network method for prediction of 2006 World Cup Football Game. In: *The 2010 International Joint Conference on Neural Networks*, Barcelona, Spain, 18-23 July. IEEE, 1-8. Available from https://ieeexplore.ieee.org/abstract/document/5596458 [accessed 15 January 2024].

Hucaljuk, J. and Rakipovic, A. (2011) Predicting football scores using machine learning techniques. In: *2011 Proceedings of the 34th International Convention MIPRO*, Opatija, Croatia, 23-27 May. IEEE, 1623-1627. Available from https://ieeexplore.ieee.org/abstract/document/5967321 [accessed 15 January 2024].

Lichess (n.d.) *Lichess.org API reference (2.0.0).* Lichess. Available from https://lichess.org/api [accessed 15 January 2024].

Pandas (2024) *pandas documentation*. Pandas. Available from https://pandas.pydata.org/docs/ [accessed 20 December 2023].

Pulido, H.A.R. (2016) *Predicting the Outcome of a Chess Game by Statistical and Machine Learning Techniques.* Master's Thesis, Universitat Politecnica de Catalunya. Available from https://upcommons.upc.edu/handle/2117/106389 [accessed 16 January 2024].

Python Software Foundation (2024. a) *json – JSON encoder and decoder*. Python Software Foundation. Available from https://docs.python.org/3.12/library/json.html [accessed 18 December 2023].

Python Software Foundation (2024. b) *urllib – URL handling modules.* Python Software Foundation. Available from https://docs.python.org/3/library/urllib.html [accessed 19 December 2023].

Python Software Foundation (2024. c) *pickle – Python object serialization.* Python Software Foundation. Available from https://docs.python.org/3/library/pickle.html [accessed 21 December 2023].

PyTorch (n.d.) *PyTorch: Key Features & Capabilities.* PyTorch. Available from https://pytorch.org/ [accessed 18 December 2023].

Reddy, K.V., Kumar, B., Kumar, N.P., Parasuraman, T., Balasubramanian, C.M. and Ramakrishnan, R. (2023) Chess Match Outcome Prediction via Sequential Data Analysis with Deep Learning. In: *2023 International Conference on Sustainable Communication Networks and Application*, Theni, India, 15-17 November. IEEE, 1442-1447. Available from https://ieeexplore.ieee.org/abstract/document/10370166 [accessed 16 January 2024].

Scikit-learn (2024) *scikit-learn: Machine Learning in Python.* Scikit-learn. Available from https://scikit-learn.org/stable/ [accessed 18 December 2023].

Somboonphokkaphan, A., Phimoltares, S. and Lursinsap, C. (2009) Tennis Winner Prediction based on Time-Series History with Neural Modeling. In: *Proceedings of the International MultiConference of Engineers and Computer Scientists 2009*, Hong Kong, 18-20 March. Hong Kong, Newswood Limited, 127-132. Available from https://www.iaeng.org/publication/IMECS2009/IMECS2009_pp127-132.pdf [accessed 16 January 2024].

TensorFlow (n.d.) *Create production-grade machine learning models with TensorFlow.* TensorFlow. Available from https://www.tensorflow.org/ [accessed 18 December 2023].

Thabtah, F., Padmavathy, A.J. and Pritchard, A. (2020) Chess Results Analysis Using Elo Measure with Machine Learning. *Journal of Information & Knowledge Management*, 19(2) 2050006. Available from https://www.worldscientific.com/doi/abs/10.1142/S0219649220500069 [accessed 17 January 2024].

Word Count

1504